

ZOL Genk

-

Operating Department



Introduction

- Critical Care Department =
 - Operating room
 - Intensive Care
 - Emergency Medicine
 - Acute Pain Service
 - Chronic Multidisciplinary Pain Centre

Operating rooms

- Campus St Jan:
 - Operating department: 18 theatres
 - Interventional department: 7 theatres
 - Ambulatory anesthesia: endoscopy, sedation (IVF, MKA)
- Campus St Barbara (Lanaken):
 - Operating department: 5 theatres
- Campus ZMK (Maaseik):
 - Operating department: 6 theatres
 - Ambulatory anesthesia: endoscopy

Operating rooms

- Total amount of procedures: > 50 000/year
- Members of staff: > 90 anesthesiologists (including trainee/fellows)
- Tertiary centre: all surgery (exc. congenital cardiac surgery and transplant surgery)
- General anesthesia/locoregional anesthesia/(local anesthesia)

Operating rooms

- All types of surgery, but also: all types of patients
 - Multiple comorbidities
 - Very young → very old
- Risk stratification

Risk stratification

- ASA: American Society of Anesthesiologist
- Category 1 to 6:

ASA Classification	Definition	Examples
ASA I	A normal healthy patient	Healthy, non-smoking, no or minimal alcohol use
ASA II	A patient with mild systemic disease	Mild diseases only without substantive functional limitations. Current smoker, social alcohol drinker, pregnancy, obesity ($30 < \text{BMI} < 40$), well-controlled DM/HTN, mild lung disease
ASA III	A patient with severe systemic disease	Substantive functional limitations; One or more moderate to severe diseases. Poorly controlled DM or HTN, COPD, morbid obesity ($\text{BMI} \geq 40$), active hepatitis, alcohol dependence or abuse, implanted pacemaker, moderate reduction of ejection fraction, ESRD undergoing regularly scheduled dialysis, history (>3 months) of MI, CVA, TIA, or CAD/stents.
ASA IV	A patient with severe systemic disease that is a constant threat to life	Recent (<3 months) MI, CVA, TIA or CAD/stents, ongoing cardiac ischemia or severe valve dysfunction, severe reduction of ejection fraction, shock, sepsis, DIC, ARD or ESRD not undergoing regularly scheduled dialysis
ASA V	A moribund patient who is not expected to survive without the operation	Ruptured abdominal/thoracic aneurysm, massive trauma, intracranial bleed with mass effect, ischemic bowel in the face of significant cardiac pathology or multiple organ/system dysfunction
ASA VI	A declared brain-dead patient whose organs are being removed for donor purposes	

Risk stratification

- ASA: American Society of Anesthesiologist
- Category 1 to 6
- Pro:
 - Easy to use
 - Limited number of categories
 - Good correlation with outcome
- Con:
 - Prone to over- and (mainly) underscoring
 - No weight adjustments to different comorbidities

Risk stratification

- Charlson Comorbidity Index (CCI)
- Lee Revised Cardiac Risk Index (RCRI)
- Frailty Score
- EuroScore-II (cardiac surgery)
- Surgical procedure risk
- ...

Risk stratification

Table 1. Charlson Comorbidity Index Scoring System

Score	Condition
1	Myocardial infarction (history, not ECG changes only) Congestive heart failure Peripheral vascular disease (includes aortic aneurysm ≥ 6 cm) Cerebrovascular disease: CVA with mild or no residua or TIA Dementia Chronic pulmonary disease Connective tissue disease Peptic ulcer disease Mild liver disease (without portal hypertension, includes chronic hepatitis) Diabetes without end-organ damage (excludes diet-controlled alone)
2	Hemiplegia Moderate or severe renal disease Diabetes with end-organ damage (retinopathy, neuropathy, nephropathy, or brittle diabetes) Tumor without metastases (exclude if >5 y from diagnosis) Leukemia (acute or chronic) Lymphoma
3	Moderate or severe liver disease
6	Metastatic solid tumor AIDS (not just HIV positive)

NOTE. For each decade > 40 years of age, a score of 1 is added to the above score.

Abbreviations: ECG, electrocardiogram; CVA, cerebrovascular accident; TIA, transient ischemic attack; AIDS, acquired immunodeficiency syndrome; HIV, human immunodeficiency virus.

Risk stratification

- Preoperative evaluation
 - Preoperative consultation with anesthesiologist
 - Specialised nurses
 - Preoperative rehabilitation
- Aim:
 - Documentation of current medical status
 - Information of surgery/type of anesthesia
 - Risk stratification ('Can patient be operated on?')
 - Predicting outcome?

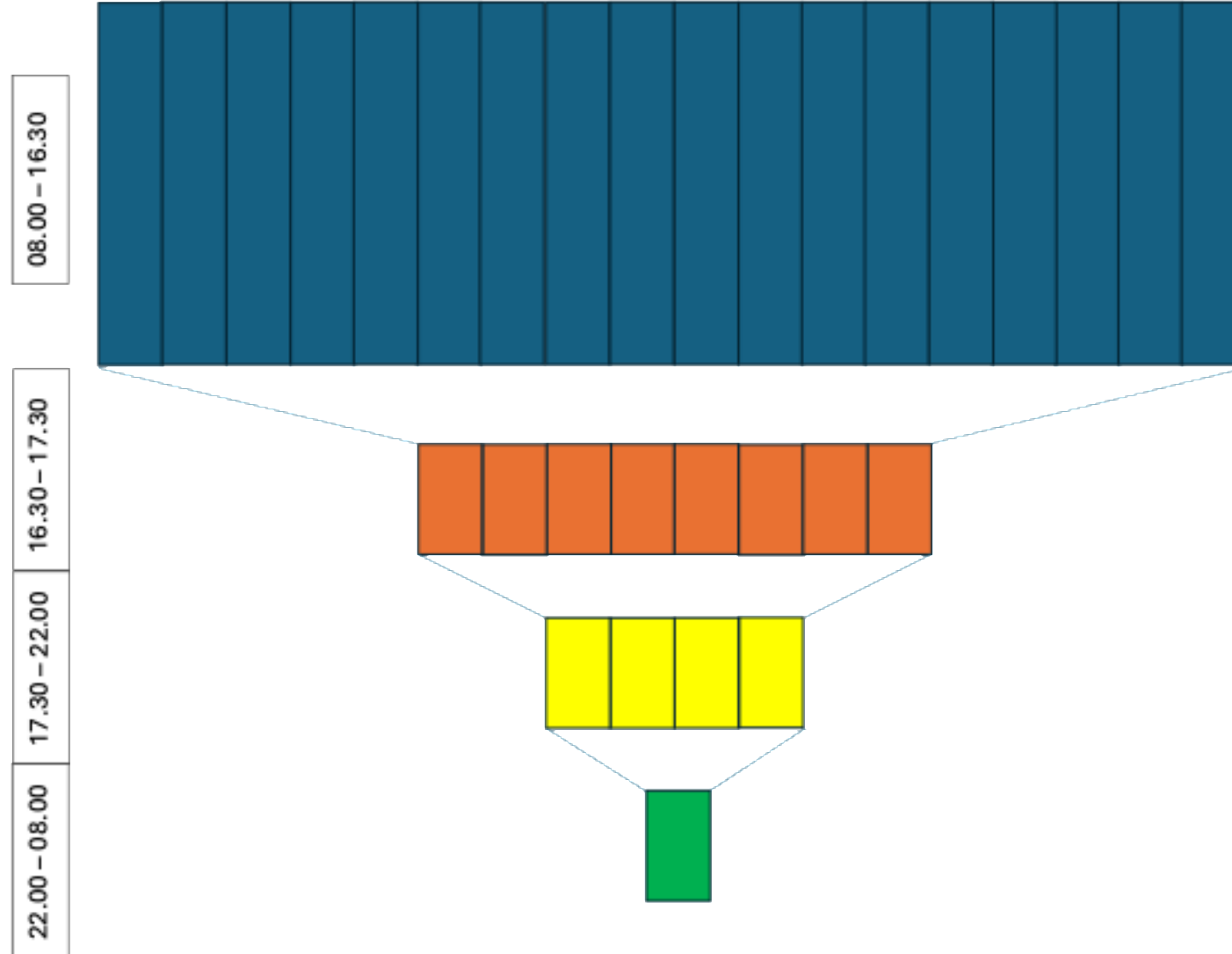
Surgical outcome

- Mortality – morbidity
- Cox Regression Analysis (proportional hazard)
- VLAD: variabel life adjustment displays
 - https://www.ucl.ac.uk/clinical-operational-research-unit/sites/clinical_operational_research_unit/files/vladmethods.pdf

Daily practice

- Daytime surgery
 - 08.00 – 16.30
 - 2-3 nurses/OR room
- On call service
 - 16.30 – 17.30: decrease to 8 OR rooms
 - 17.30 – 22.00: decrease to 4 OR rooms
 - 22.00 – 08.00: 3 nurses on call (max. 1 OR room)
- Back-up nurses
 - Available in night time service (22.00 – 08.00) and weekend/bank holidays 24/7

Daily practice



Luchthavenbord

Tiidsinterval 30 minuten < 29-01-2025 > woensdag 29 januari 2025 (Let op! Dit is niet vandaag!)

	GO01	GO02	GO03	GO04	GO05	GO06	GO07	GO08	GO09	GO10	GO11	GO12	GO13	GO14	GO15	GO16	GO17	GO18	KO05
08:00	IC 1 OSTEOP MEYNJ Scr aldc K	1 REDUC VERMB Scr aldc K	1 LAP,AD DEJOE Scr aldc K	1 LAP,HE KNOLJ Scr aldc K	ROB... 1 GASTRI BOUCK Scr aldc K	1 RAL PA BYNEB Scr aldc K	1 SECTIO NULEK Scr aldc K	1 CROSS LANSW Scr aldc K	1 CAROTI NAUWS Scr aldc K	3 CORAC SCHIS Scr aldc K	1 TONGI CORNE Scr aldc K	1 TREPAN OOSTC Scr aldc K	1 VENTRI CORNE Scr aldc K		1 CLAVIC OPSOG Scr aldc K	1 UNICOI OOSTJ Scr aldc K	1 ZENUW Scr aldc K	1 TENI RA Scr aldc K	1 gefene LANSW Scr aldc K
09:00																			
10:00																			
11:00	1 KLEINE MEYNJ Scr aldc K	1 HEELKL VERMB Scr aldc K	1 MASTE DEJOE Scr aldc K	1 LAP,HE KNOLJ Scr aldc K	1 LAP,GA BOUCK Scr aldc K	1 RAL NE BYNEB Scr aldc K	1 SECTIO KERKC Scr aldc K	1 AV FIST NAUWS Scr aldc K					1 THORA CORNE Scr aldc K	1 HUIDLE MENTV Scr aldc K	1 SCHOU OPSOG Scr aldc K	1 TOTALE OOSTJ Scr aldc K	1 DUIMPI DUERJ Scr aldc K	1 DUIMPI DUERJ Scr aldc K	1 gefene LANSW Scr aldc K
12:00																			
13:00	1 EX Scr aldc K	1 REDUC VERMB Scr aldc K		1 LAP,GA KNOLJ Scr aldc K	1 GASTRI BOUCK Scr aldc K	1 RAL PA BYNEB Scr aldc K	1 SECTIO KERKC Scr aldc K	1 WOND LANSW Scr aldc K	1 AV FIST NAUWS Scr aldc K	1 EXTRAC Scr aldc K	1 EXPLOF WICH Scr aldc K			1 HEMI-T MENTK Scr aldc K	1 ENKEL OPSOG Scr aldc K	1 TOTALE OOSTJ Scr aldc K	1 DUIMPI DUERJ Scr aldc K	1 DUIMPI DUERJ Scr aldc K	1 gefene LANSW Scr aldc K
14:00	1 EXTRAC Scr aldc K		1 MASTE DEJOE Scr aldc K		1 GASTRI BOUCK Scr aldc K	1 RAL PA BYNEB Scr aldc K	1 SECTIO KERKC Scr aldc K	1 AMPUT NAUWS Scr aldc K	1 AV FIST NAUWS Scr aldc K	1 EXTRAC Scr aldc K	1 EXPLOF WICH Scr aldc K			1 HEMI-T MENTK Scr aldc K	1 ENKEL OPSOG Scr aldc K	1 TOTALE OOSTJ Scr aldc K	1 VINGER DUERJ Scr aldc K	1 VINGER DUERJ Scr aldc K	1 DILATA LANSW Scr aldc K
15:00	1 EXTRAC Scr aldc K	1 RECON VERMB Scr aldc K		1 ILEOSTI KNOLJ Scr aldc K	1 EXPLOF Scr aldc K														
16:00	1 FRACTU Scr aldc K																		
17:00																			
18:00					1 LAP,GA WICH Scr aldc K				1 AV FIST NAUWS Scr aldc K										
19:00					1 LAP,GA WICH Scr aldc K						1 HEUP F VANDF Scr aldc K								
20:00																			
21:00					1 EXPLOF WICH Scr aldc K														
22:00																			

Dagmelding

EX-OK G... v

Spoegevalen

- SPG - Rode kant
- SPG - Blauwe kant A
- SPG - Blauwe kant B
- SPG - Groene kant
- SPG CEDOCS SJ
- Dagoverzicht SEH

Luchthavenbord

- Luchthavenbord
- Luchthavenbord K-Blok
- Luchthavenbord Maaseik
- Luchthavenbord SB

OK Genk

- Map OKC SAS Genk
- Map OKC Operatiekamers Genk
- Map OKC Grote Recovery Genk
- Map OKC Kleine Recovery Genk
- Map OKC SAS Genk Intervention...
- Map OKC Operatiekamers Genk...
- Map OKC Kleine Recovery Genk Interv...

Daily practice

- Time registration:
 - Nurse-based for theatre in, doctor-based for theatre out
 - Room in and room out are the only reliable data available!
 - Other data registration:
 - Called for patient
 - Patient in operating department
 - Start induction – end induction
 - Start operation
 - End operation
 - Patient in recovery
 - Patient to ward

PDMS OK

Zaal 13 (inzage) De Deyne, Cathy

Operatie - Wijzigen

Operatiekamer

GO13

Zaal 13

Bed

GO13

Aanvrager

118395

Engelborghs, Koen

NCH

Operateur

118395

Engelborghs, Koen

NCH

Behandeling

HYPOFYSETUMOR TRANSSPHENOIDAAL ENDOSCOPISCH

Zijde

-

n.v.t.

Opmerking

Prioriteit

Hoofd anesthesietechniek

Sub anesthesietechniek

Ligging

ASA

3

Ernstige functiebeperkende systeemafwijking

Postop. bestemming

IZ

IZ

Traject

Operatiernr.

0100761389

Operatiememo

transsfenoid hypofyse

Soort operatie

Operatietijden

Gebeld

< 10:58 >

Op sas

< 11:19 >

Start locoregionaal

< _ _ >

Eind locoregionaal

< _ _ >

Klaar voor OK

< 11:19 >

Op OK

< 11:37 >

Start inleiding

< 11:50 >

Eind inleiding

< 11:50 >

Start snijder

< _ _ >

Eind snijder

< _ _ >

Uit OK

< _ _ >

Tijd recovery

< _ _ >

Bellen afdeling

< _ _ >

Naar afdeling

< _ _ >

NCH

Engelborghs, Koen

1

HYPTUMTS

HYPOFYSETUMOR TRANSSPHENOIDAAL ENDOSCOPISCH

Overige operateurs

Naam	Spec.	Artstype
Cicek, Abdulhamid	NCH	Assistent

Anesthesiologen/SPS

Naam	Artstype	Aflos	Start	Eind	Supervisor
De Deyne, Cathy	Interne specialist		:	:	

Teamleden

Voornaam	Naam	Functie	Aflos	Start	Eind	Opmerking
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Artikelen

Status Imp.	Omschrijving	Artikelcode	KZP	Scancode(s)
	INSTILLAGEL SPIJT 11 ML	1409508	HYPTUMTS_118395	49742/1409508
	ISO-BETADINE DERMICUM OPL 125 ML	1422188	HYPTUMTS_118395	1422188
	CEDIUM CHLORHEXIDINI 0,05% UNIT-DOSE 10 ML	436395	HYPTUMTS_118395	2842946/436395/2030004363953
	MINI-PLASCO WATER AMP 20 ML MP60	4754116	HYPTUMTS_118395	04030539076234/4030539104661/819128/4754116
	NACL 0,9% FRESINIUS VERSYLENE GIETFLES 500 ML	4802212	HYPTUMTS_118395	1509314/4802212
	ALU GODET 251ML STER.LAM REF.3027648	SM00078	HYPTUMTS_118395	SM00078
	MICROCHIRURISCH LEMMET 15" Ref. 72-1532	SO00167	HYPTUMTS_118395	SO00167
	BISTOURIMESJE NR 23 REF.19-0510223 Ref. 19-0510223	SO00191	HYPTUMTS_118395	SO00191
	DIATHERM.ZAKJE 40X35CM REF.707035 Ref. 707035	SO00687	HYPTUMTS_118395	SO00687
	HANDSCHOENEN ETHIPARAT M M3345	ZH00043257	HYPTUMTS_118395	9007676
	HEIDELBERG VERLENGSTUK 75CM Ref. 4097173	SI00842	HYPTUMTS_118395	SI00842
	SURGICAL PATTIE 13X13MM 10 DR. Ref. H4B21A805	SO01053	HYPTUMTS_118395	SO01053
	SURGICAL PATTIE 13X51MM 10 DR. 80-1406	ZH00043500	HYPTUMTS_118395	9010555
	VICRYL RAP UND 75CM FS-2 3-0 (M2) VR2298	9012775	HYPTUMTS_118395	2030090127750/9012775
	VICRYL PLUS UND 70CM FS-1 3/0 M2 VCP442H	9031205	HYPTUMTS_118395	2030090312057/9031205
	TEGADERM TRANSP.FILMVERBAND10X12CM-1626W	SW30254	HYPTUMTS_118395	SW30254
	TROUSS INTRAFIX SAFESET+ INSPIJTPUNT180CM REF.4063005	SI03287	HYPTUMTS_118395	SI03287
	POTJE STERIEL 60ML	SM03345	HYPTUMTS_118395	
	HANDSCH. GAMMEX LATEX TEXTURED 6,5 Ref. 331300665	SP05664	HYPTUMTS_118395	SP05664
	HANDSCH. GAMMEX LATEX TEXTURED 7,5 Ref. 331300675	SP05666	HYPTUMTS_118395	SP05666
	BIPOLAR CORD DISP, UNITISED PLUG 3.0M S Ref. 801188	SO04402	HYPTUMTS_118395	SO04402
	SONDAGESET B1267 REF.4783641 Ref. 4783642	SU05027	HYPTUMTS_118395	SU05027
	CAMERA DRAPE (MICROTEC)9900UK 9900UK	ZH00046848	HYPTUMTS_118395	9100026

Financiële afhandeling

Ingeleugde gebruiker...

Traceerbaar

Halfvol

Voorschrift

Prijs

Opname koppelen

Opname inzien

Vertraging

Dag eerder

Dag later

Detail

Overige acties

Opslaan

OK

Annuleren

nhuis
imburg

Daily practice

- Planning of surgery:
 - Surgeon (or representative) puts in request for surgery
 - Automated lists, corresponds with:
 - Procedure
 - Acting surgeon
 - Length of surgery
 - Materials (used by storage to prepare case cart)
 - Patient variables:
 - Priority (elective, urgent, emergent, salvage)
 - Hospitalisation/day care surgery
 - ASA classification
 - (anticoagulant drug management)

Van Bylen Ben - IZ - HiX_PRODUCTIE - 6.2 HF147.1 - HiX - ChipSoft

Gastro-enterologische chirurgie
23-08-2021 : Gastro-enterologische chirurgie

Order plaatsen - Aanvraag Opname + OK - plannen

Aanvragend arts: 173880 K. Govaerts (Interne specialist, GOC)

Aanvragende locatie: Sint-Jan, Genk Aanvragende afdeling: Intensive Care

Setting: Poliklinisch

Behandelaar: 173880 Govaerts, Kim GOC

Operateur: 173880 Govaerts, Kim GOC

Behandelgegevens

Zoek behandeling: LAPGAL ☒ + OK

Vrije tekst behandeling: LAPGAL Zijde: - n.v.t.

Operatiegegevens

Prioriteit: ASA: 3 Ernstige functiet

Geplande sessie: <-- dubbel klik Operatielocatie: Sint-Jan, Genk

Zaal, datum/tijd: Operatieduur: 71

Anesthesie- & subtechniek:

Nuchterbeleid: Houding:

Postop. bestemming: Duur:

Opnamegegevens

Opnametype & -subtype: Dagziekenhuis /

Opnameafdeling: Opnamelocatie: G Sint-Jan, Genk

Pre- & postop. dagen: 0 - + / 0 - + Opnameduur: 1 D

Kamertype gevraagd: Reden:

Resources

☐ Bloedleegte ☐ Cellsaver ☐ Scopie ☐ Microscop ☐ Leeninstrumentarium (specificeer in operatiememo)

Planningsmemo Opnamememo Operatiememo

Stopmoment Resources Verrichtingen

Auteur: Van Bylen Ben (31-01-2025 12:58) ☐ Definitief

Stopmoment 1a - Spreekuur behandelaar Antico-antiaggregantia beleid Medicatie beleid bij ontslag op dagziekenhuis Scorelijsten

Stopmoment 1a - Spreekuur behandelaar

• Stopmoment 1a niet gevuld i.v.m. livegang ☐ Ja ☒ Nee

OK Nieuwe operatie... Andere operatie Detail Overige acties OK Annulleren

To be investigated: efficiency

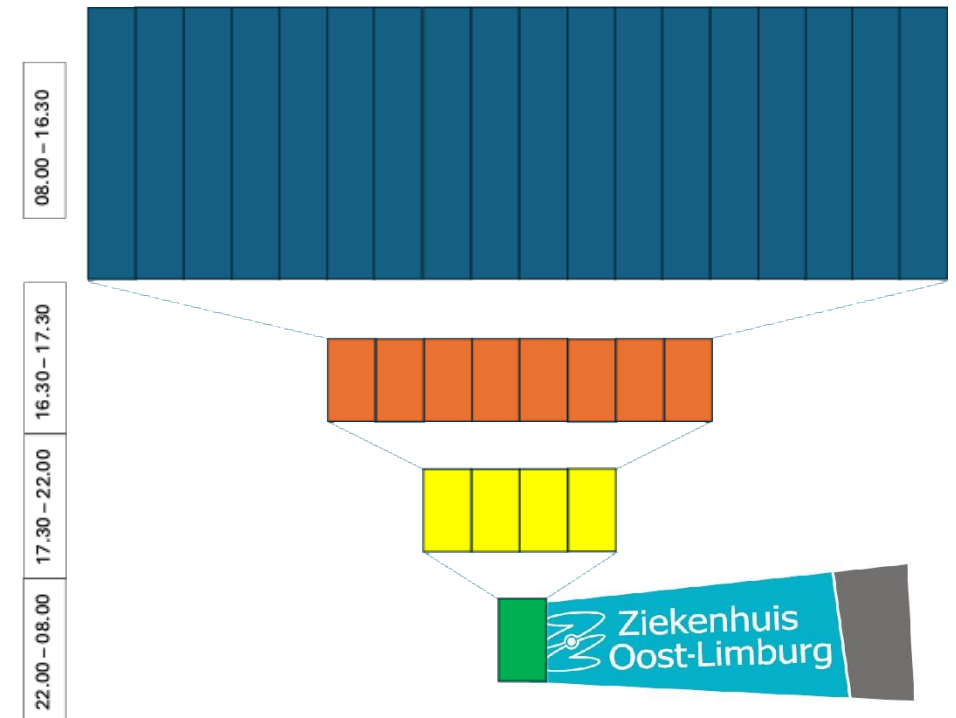
- Does scheduled OR time reflect observed OR time?
- Does observed OR time correspond with current working schedule?
 - Determining area under the curve
- Can we improve OR scheduling efficiency?
 - Efficient use of emergency theatre?
 - Creating gaps for non-elective surgery?
- Can we predict OR scheduling?
 - Using patient and surgery related variables?

To be investigated: safety

- Does predicted mortality correspond with observed mortality?
- Does OR staffing/efficient use of OR time correspond with mortality/outcome?
 - Linking AUC and mortality?
- Can we predict mortality more accurate using patient and surgery variables?

To be investigated

- Phase 1:
 - Retrospective analysis of timing scheduled surgery versus reality
 - Using existing data
 - Calculating area under the curve
 - Q:
 - Does scheduled operating time corresponds with real situation?
 - Added value of 'swapping theatres?'



To be investigated

- Phase 2:
 - Prospective study, incorporating patient variables (CCI, ASA classification, type of surgery...)
 - Does predicted mortality reflect observed mortality?
 - Using VLAD curves
- Phase 3:
 - Prediction of operating room scheduling